

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course : Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVOE605H - APPLICATIONS OF REMOTE SENSING AND GEOGRAPHIC INFORMATION SYSTEMS

Time : 1.5 hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1. Which of the following best defines "remote sensing"?

- a. Collecting data by physically touching the object
- b. Collecting information about an object without being in direct contact
- c. Only using satellites to collect data
- d. Capturing images through a camera

Which part of the electromagnetic spectrum is commonly used in optical remote sensing?

- a. X-rays
- b. Ultraviolet
- c. Visible and Infrared
- d. Microwave

3. What does "spatial resolution" in remote sensing refer to?

- a. The ability to generate 3D models
- b. The distance a satellite can travel in one orbit
- c. The smallest detail that can be detected by the sensor
- d. The altitude of the satellite

Why might satellite remote sensing be preferred over ground surveying in highway planning?

- a. It is less accurate
- b. It covers smaller areas
- c. It provides fast, large-area data collection
- d. It is dependent on direct contact

5. What is the primary purpose of image enhancement in remote sensing?

- a. To decrease spatial resolution
- b. To increase cloud cover
- c. To improve image interpretability
- d. To compress file size

- 6 In the context of visual interpretation, “shadow” helps to identify which of the following features best?
- Soil texture
 - Elevation of objects
 - Water content
 - Color variation
- 7 Which file format is commonly used for vector data?

- GeoTIFF
- Shapefile
- PNG
- CSV

What does GIS stand for?

- Global Imaging System
- Geographic Information System
- Geo-based Interaction Software
- General Image Storage

- 9 What is the main purpose of geo-referencing in GIS?

- To print maps on paper
- To increase map resolution
- To assign real-world coordinates to a digital image or map
- To compress spatial data files

What is the shape of the Earth typically approximated as in most geographic coordinate systems?

- Perfect sphere
- Flat plane
- Ellipsoid
- Cube

- 11 Which of the following is a datum used in GIS?

- Google Earth
- WGS 84
- ArcGIS
- Vector Map

Why are map projections necessary?

- To calculate internet speed
- To store raster files
- To represent the 3D Earth on 2D surfaces like maps
- To enhance satellite imagery

Q2. Solve the following.

- What are commonly used websites or platforms to download remote sensing data? 6
- Explain various important elements of visual interpretation of image Data with neat labelled diagram. 6

Q3. Solve the following.

- A) Explain the ideal and real remote sensing systems in terms of all remote sensing concepts. 6
- B) Describe how LiDAR works and mention its key benefits and limitations. 6
- Q4. Solve Any Two of the following.
- A) Compare supervised and unsupervised classification techniques with advantages and limitations. 6
- B) Describe various thematic maps generated through remote sensing and civil engineering applications. 6
- C) Evaluate the applications of remote sensing and GIS in water resource and watershed management. 6
- Q5. Solve Any Two of the following.
- A) Compare raster and vector data formats with examples of their use in real world GIS applications. 6
- B) Describe the differences between georeferencing and geocoding in GIS applications. 6
- C) Write a detailed note on Geographic Information System with its advantages and limitations. 6
- Q6. Solve Any Two of the following.
- A) What is a datum? Explain different types of datum systems used in geodesy. 6
- B) Differentiate between DEM, DTM, and DSM with neat diagrams and applications. 6
- C) Explain the concept of geographic coordinate systems with the help of a diagram, including its strengths and weaknesses. 6

*** End ***